

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1255

Isabella 100 kV
February 24, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1255. The Interconnection Customer proposes a 108.9 MW solar photovoltaic facility interconnecting at the Isabella 100 kV substation in the Ohio area. The request was received on July 20, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 388 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,158,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
E Cortez / John C. Eisinger 100kV	Omni-Hackberry / Norwich 500kV	134.5	126	14.1
Jug Street / Goose Lake 100kV	Omni-Hackberry / Waynoka Gas Processing Plant 230kV	118.8	157	8.5
Mead Pump / Goose Lake 100kV	Waynoka Gas Processing Plant / Omni-Hackberry 230kV line	118.0	141	8.5
Mead Pump / Magdalena 100kV	Waynoka Gas Processing Plant / Omni-Hackberry 230kV Line	112.1	132	8.5
Magdalena / Daly City Tap 100kV Line	Omni-Hackberry / Waynoka Gas Processing Plant 230kV	106.3	125	8.5
Penryn Substation / John C. Eisinger 100kV	Denmark Substation / Thrive Fontana 100kV	105.7	133	8.6
Opp Substation / Penryn 100kV	Denmark / Thrive Fontana 100kV line	100.2	128	8.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
John C. Eisinger / East Cortez 100kV line	Rebuild line with higher-capacity conductor	\$8,370,000	100% (\$8,370,000)
Goose Lke / Jug St Substation 100kV	Rebuild line with higher-capacity conductor	\$1,601,000	100% (\$1,601,000)
Mead Pump / Goose Lake 100kV L	Rebuild line with higher-capacity conductor	\$1,232,000	100% (\$1,232,000)
Mead Pump / Magdalena 100kV	Rebuild line with higher-capacity conductor	\$1,711,000	100% (\$1,711,000)
Daly City Tap / Magdalena 100kV	Reconductor line	\$1,233,000	100% (\$1,233,000)
Penryn / John C. Eisinger 100kV L	Reconductor line	\$1,182,000	100% (\$1,182,000)
Opp / Penryn 100kV	Reconductor line	\$829,000	100% (\$829,000)
	Total allocated to project		\$16,158,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.